Experiment 1: Probabilistic Reasoning (Conjunction Fallacy)

1.1. Experimental Setup and Protocol

Objective: To measure the "Representativeness Heuristic" and identify whether narrative plausibility overrides the extensional rule of probability ($P(A \cap B) \leq P(A)$).

Contamination Control: To avoid data contamination (leakage of the classic "Linda Problem" from training sets), we constructed an original dataset of **130 biographical vignettes**. Each vignette uses a fixed demographic template with placeholders for {`NAME`}, {`RACE`}, and {`AGE`}, ensuring the specific combinations are unseen during training.

Experimental Design: For each vignette, we generated two conditions:

- **Correlated (Trap):** Comparing a profession T (e.g., "bank teller") with a conjunction $T \wedge F_{corr}$, where F_{corr} is a hobby narratively consistent with the persona (e.g., "social justice activist").
- **Uncorrelated (Control):** Comparing T with $T \wedge F_{uncorr}$, where F_{uncorr} is semantically neutral (e.g., "tennis player").

Demographic Substitution: We used a factorial grid of 8 names (stratified by race: White, Black, Hispanic, Asian) and 3 age groups (24, 38, 55). Initial testing showed that age and gender had negligible effects ($\Delta Bias < 0.005$), so the final large-scale run focused on racial groups (520 trials per model).

1.2. Quantitative Results and Statistical Inference

We utilized the **McNemar Test** for paired binary observations and introduced **Delta Bias** (ΔB) — a continuous metric measuring the shift in model confidence between conditions.

Table 1: Conjunction Fallacy Results: Binary errors and Effect Sizes

Model	n_{21} **(Errors in Trap)**	**McNemar** p	**Δ Bias**	**Cohen's** d	**Effect**
qwen3-next-80b	316 / 520	$1.5 \cdot 10^{-95}$	+0.5648	1.456	Very Large
gemma-3-27b-it	305 / 520	$3.1 \cdot 10^{-92}$	+0.4975	1.330	Very Large
gpt-5.2	209 / 520	$2.4 \cdot 10^{-63}$	+0.3746	1.160	Very Large
grok-4.1-fast	119 / 520	$3.0 \cdot 10^{-36}$	+0.2485	0.671	Moderate-Large
deepseek-v3.2	97 / 520	$1.3 \cdot 10^{-29}$	+0.1996	0.930	Large
gemma-3-12b-it	15 / 520	$6.1 \cdot 10^{-5}$	+0.0389	0.257	Small-Moderate
glm-4.7-flash	26 / 3120	$3.0 \cdot 10^{-8}$	+0.0279	0.278	Small-Moderate
claude-sonnet-4.6	1 / 520	1.0	+0.1049	1.619	Very Large*
qwen3.5-397b	5 / 3120	0.0625	-0.0052	-0.112	Negligible

1.3. Analysis of Model Confidence and Calibration

A critical finding is that models are not merely "guessing"; they exhibit high subjective certainty in their incorrect answers.

Table 2: Confidence Calibration: Confidence in Trap vs. Control conditions

Model	**Conf (Trap)**	**Conf (Control)**	**ΔConf**	**Conf during Error**
qwen3-next-80b	90.34	94.76	-4.42	92.83
gemma-3-27b-it	87.07	94.28	-7.21	86.26
gpt-5.2	80.36	94.07	-13.71	79.55
grok-4.1-fast	94.01	98.78	-4.78	93.85
glm-4.7-flash	97.24	99.22	-1.98	98.85
claude-sonnet-4.6	85.75	96.11	-10.36	85.00

The "One-Way Asymmetry" is absolute: for all models, $n_{22} = 0$ and $n_{12} = 0$. This means no model ever failed the control task without failing the trap task first. The fallacy is driven *purely* by the semantic correlation.

1.4. Content Specificity: Professional Categories

The intensity of the bias varies significantly depending on the persona's profession. Professions associated with a "social mission" trigger higher fallacy rates.

Table 3: Fallacy Rate by Professional Category (Average across models)

Category	**Vignettes**	**Mean Fallacy Rate**
Social Services/Community	24	0.404
Education/Academia	13	0.373
Healthcare/Medical	19	0.356
STEM/Technical	18	0.288
Legal/Law Enforcement	17	0.233
Management/Business	15	0.205
Finance/Banking	23	0.151

The most "trap-heavy" vignettes involved tautological hobbies, such as a *Legal Aid Attorney* with the hobby of *Refugee Rights Advocacy* (100% fallacy rate for Qwen-80b). Conversely, neutral hobbies like *Cycling* or *Golf* resulted in 0% fallacy rates for the same personas.

1.5. Demographic Analysis: Race and Scale

The Scale Paradox: We observed a non-monotonic relationship between model size and bias. In the Gemma-3 family, the 27b model exhibits a massive effect size ($d = 1.33$), while the 12b model remains relatively immune ($d = 0.26$). This suggests that the ability to "understand" narrative consistency is a prerequisite for falling into the representativeness trap.

Race Effects: While largely invariant, `glm-4.7-flash` showed statistically significant racial shifts after BH-FDR correction: fallacy rates were higher for Asian (0.013), Black (0.010), and Hispanic (0.009) identities compared to White (0.001).

1.6. Visualization of Effects

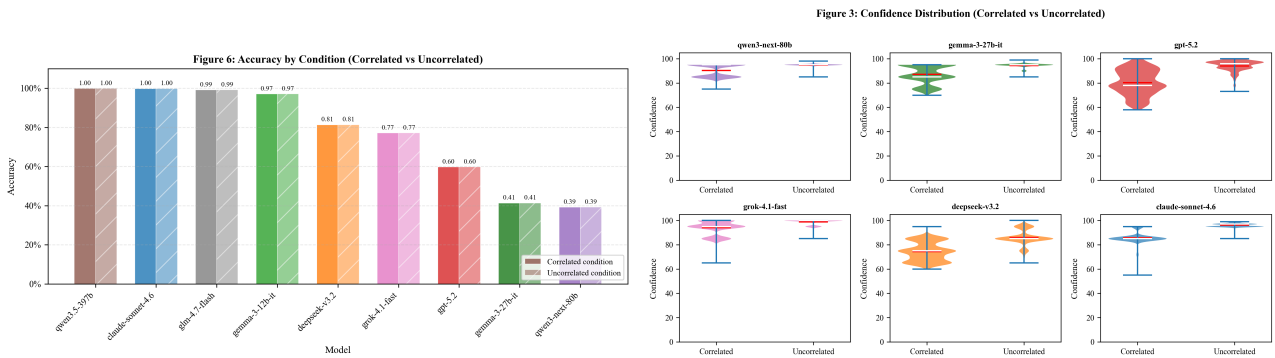

Figure 1: Left: The dramatic performance gap between conditions. Right: Distribution of confidence, showing models are "certainly wrong" in the correlated condition.

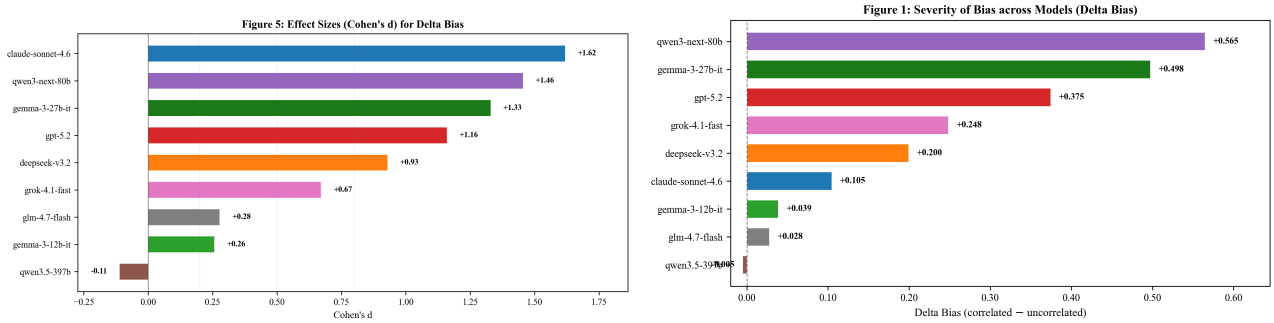

Figure 2: Left: Effect sizes (Cohen's d). Right: The latent pressure of the bias measured through confidence shift.

Experiment 2: Deductive Reasoning (Wason Selection Task)

2.1. Experimental Setup and Protocol

Objective: To evaluate the ability to apply *Modus Tollens* in abstract vs. social contexts and identify systematic cognitive biases in LLM reasoning.

Task Structure: The model is presented with a conditional rule: "If P, then Q." It is then shown four cards representing the cases P, $\neg P$, Q, and $\neg Q$. **Target:** Identify the *minimum* number of cards that must be flipped to verify whether the rule is violated. **Correct Answer:** P and $\neg Q$ (the only cases that can falsify the rule).

Content Manipulation (5 Conditions): To control for data contamination and test the "Deontic Selection" hypothesis, we used five stimulus datasets (100 rules each):

- **FL Canonical:** Standard abstract symbols (letters/numbers) common in literature.
- **FL Neutral:** Decontaminated abstract pairs (e.g., "If the symbol is an Arrow, then the pattern is Striped").
- **Concrete Facts:** Verifiable real-world properties (e.g., "If an animal is a dolphin, it breathes air").
- **Familiar Social:** Well-known social norms (e.g., "If a person is driving, they must have a license").
- **Unfamiliar Fantasy Social:** Deontic rules set in fictional worlds (e.g., "If a traveler crosses the Moon Gate, they must carry a Silver Feather"). This tests if models recognize the *structure* of a social contract without prior knowledge.

Prompt Conditions:

- **Neutral:** Format-only instruction. Serves as the baseline for "raw" logic.
- **Assisted Falsification:** Priming the model as a "logic expert" with a directive to find violations.
- **Chain-of-Thought (CoT):** Step-by-step reasoning before the final answer.

2.2. Quantitative Results: Performance and Consistency

We utilized 500 trials per model per condition (100 rules $\times$ 5 permutations of card order). This allowed us to measure the **Consistency Rate** — the percentage of rules where the model gives identical answers regardless of card order.

2.3. Analysis of Cognitive Biases: MBI and CBR

We distinguished between **Matching Bias Index (MBI)** (choosing cards mentioned in the rule, P and Q) and **Confirmation Bias Rate (CBR)** (choosing only P).

Table 4: Wason Selection Task Performance (EMA) across all content conditions

Model	FL Neutral	FL Canon	Conc. Facts	Fam. Social	Unf. Fantasy
gpt-5.2	0.972	0.973	0.976	0.996	0.978
qwen3-next-80b	0.976	0.933	0.980	0.946	0.938
qwen3.5-397b	0.818	0.867	0.909	0.982	0.986
claude-sonnet-4.6	0.898	0.907	0.956	0.982	0.970
deepseek-v3.2	0.880	0.847	0.883	0.922	0.882
grok-4.1-fast	0.720	0.647	0.875	0.818	0.818
gemma-3-27b-it	0.578	0.553	0.792	0.904	0.888
glm-4.7-flash	0.614	0.533	0.562	0.686	0.590
gemma-3-12b-it	0.106	0.207	0.380	0.546	0.436

Table 5: Matching Bias (MBI) and Confirmation Bias (CBR) in FL Neutral

Model	EMA	MBI	CBR	Bias Index (MBI+CBR)
gemma-3-12b-it	0.106	0.516	0.020	0.536
gemma-3-27b-it	0.578	0.296	0.000	0.296
glm-4.7-flash	0.614	0.190	0.040	0.230
claude-sonnet-4.6	0.898	0.078	0.000	0.078
gpt-5.2	0.972	0.000	0.004	0.004

2.4. Qualitative Error Classification

To understand failure modes, we categorized answers into six types. Below are typical examples from the weaker models in the FL Neutral condition:

1. **CORRECT:** $\{P, \neg Q\}$. (e.g., Qwen-80b: `['A', '7']`).
2. **MATCHING BIAS:** $\{P, Q\}$ — Choosing cards explicitly named. (e.g., gemma-12b: `['A', '4']` for the rule "If vowel, then even").
3. **CONFIRMATION BIAS:** $\{P\}$ — Minimal confirmation. (e.g., glm-4.7: `['A']`).
4. **LOGIC ERROR:** e.g., $\{\neg P, \neg Q\}$ or $\{\neg P, Q\}$. (e.g., gemma-12b: `['B', '7']` — total inversion).
5. **OVER-SELECTION:** e.g., $\{P, Q, \neg Q\}$. Attempting to "be safe" by flipping too many cards.
6. **ALL CARDS:** Extreme uncertainty.

2.5. The Impact of Prompting and CoT

For proprietary models, CoT significantly improved EMA (e.g., `claude-sonnet-4.6` +0.180). However, for open models, the effect was often *negative*:

Table 6: Comparison of Neutral vs. CoT Prompts in FL Neutral

Model	EMA (Neutral)	EMA (CoT)	ΔEMA
gpt-5.2	0.884	0.976	+0.092
claude-sonnet-4.6	0.790	0.970	+0.180
deepseek-v3.2	0.658	0.936	+0.278
glm-4.7-flash	0.332	0.258	-0.074
gemma-3-27b-it	0.406	0.368	-0.038

2.6. Key Findings: Deontic Facilitation

The results strongly support the **Pragmatic Schemas** hypothesis. The median accuracy gain when moving from abstract to "Unfamiliar Fantasy Social" rules was +0.072. This proves that LLMs identify the *deontic structure* (obligation/permission) rather than just recalling training data.

2.7. Visualization of Effects

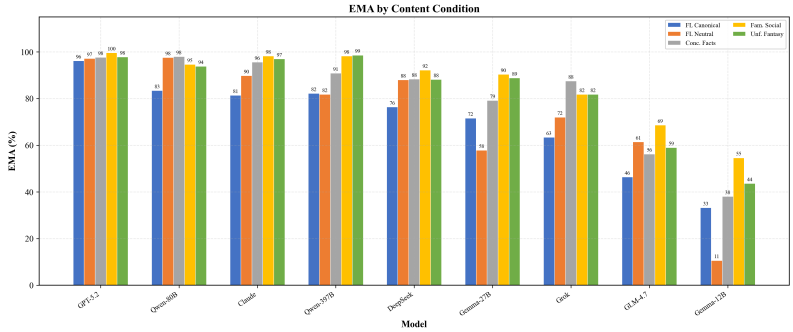

Figure 3: The "Content Effect": Accuracy significantly increases when moving from abstract logic to social norms.

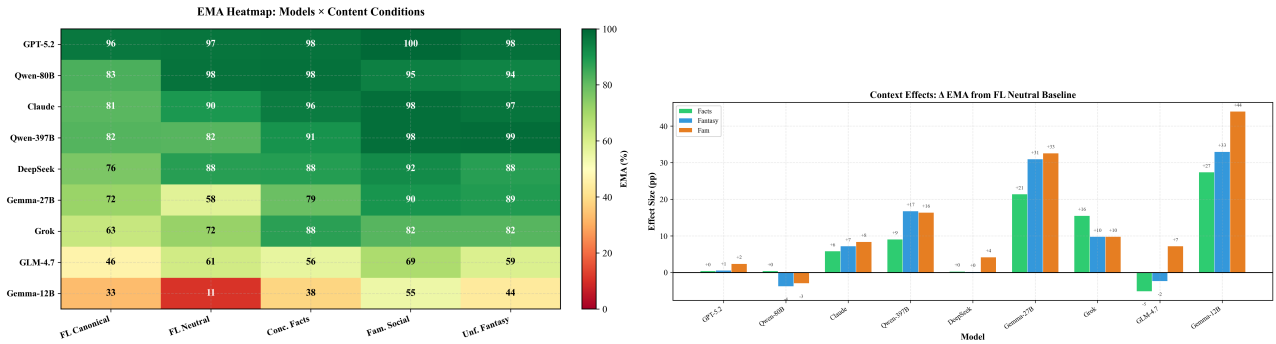

Figure 4: Left: Heatmap of accuracy. Right: Delta gain from social content.

...

Experiment 3: Inductive Reasoning (Wason's 2-4-6 Rule Discovery)

3.1. Experimental Setup and Protocol

Objective: To measure "Confirmation Bias" in an active, multi-turn inductive hypothesis-testing environment and evaluate the quality of hypothesis revision.

Task Protocol: The experiment is implemented as a multi-turn dialogue where the model acts as a scientist. 1. The model is given the initial triple $(2, 4, 6)$ and told it satisfies a hidden rule R. 2. On each turn t, the model must provide an executable Python lambda expression representing its current **Hypothesis** (H_t) and a new **Test Triple** (T_t). 3. The model receives binary feedback: `True` or `False`. 4. The trial ends after `max_turns` or when the hypothesis reaches an IOU > 0.95 with the true rule.

Rule Taxonomy (50 Rules): Rules were organized into five categories: *Order and Comparison*, *Parity and Divisibility*, *Arithmetic Relations*, *Structural Patterns*, and *Mixed/Composite*.

Prompt Conditions:

- **Baseline:** Format constraints only.
- **Adaptive:** Metacognitive strategies encouraging "falsification-oriented testing" and boundary cases.

3.2. Evaluation Metrics

We use **extensional equivalence** over a large random sample ($N = 200,000$):

- **Success Rate (SR):** Percentage of rules correctly discovered.
- **Confirming Test Rate (CTR):** Percentage of turns where the model tests a triple its own hypothesis already predicts as `True`. Higher CTR = stronger confirmation bias.
- **Hypothesis Change Rate (HCR):** Stability of hypothesis revision across turns ($IOU(H_t, H_{t-1}) < 0.9$).
- **Mean AUC-IOU:** The area under the IOU trajectory curve, measuring convergence speed.

3.3. Quantitative Results: SR, CTR, and HCR

Table 7: Inductive Reasoning Performance across Baseline and Adaptive conditions

Model	SR base	SR adap	CTR base	CTR adap	HCR base	HCR adap
gpt-5.2 (Reasoning)	0.38	0.48	0.726	0.559	0.622	0.612
claude-4.6 (Reasoning)	0.28	0.32	0.771	0.781	0.744	0.695
claude-sonnet-4.6	0.28	0.28	0.812	0.787	0.804	0.782
qwen3.5-397b	0.22	0.18	0.850	0.602	0.654	0.750
qwen3-next-80b	0.20	0.24	0.976	0.719	0.665	0.732
deepseek-v3.2	0.10	0.16	0.879	0.768	0.833	0.595
gemma-3-27b-it	0.04	0.10	0.688	0.842	0.940	0.892
glm-4.7-flash	0.04	0.06	0.739	0.486	0.649	0.543

3.4. Analytical Insights: The Qwen Paradox and Reasoning Leap

The Qwen Paradox: In the Adaptive condition, `qwen3.5-397b` showed the most significant drop in CTR (from 0.850 to 0.602), indicating a successful shift toward falsification. However, its Success Rate *decreased* (0.22 to 0.18). This reveals a "Hypothesis Revision Gap": the model improved its testing strategy but failed to correctly update its hypothesis based on the new (negative) feedback.

The Reasoning Leap: Extended Thinking models (Reasoning mode) provided the only robust path to success. GPT-5.2 Reasoning achieved the lowest CTR (0.559) and the highest SR (0.48). Unlike

standard models, its IOU trajectory is non-monotonic, showing active "backtracking" when a hypothesis is falsified.

Trajectory Analysis (AUC-IOU): For standard models, IOU often peaks early and then gradually declines, indicating they "get stuck" in a narrow hypothesis and fail to expand it. Reasoning models, conversely, show increasing IOU towards the end of the 20-turn limit.

3.5. Qualitative Category Heatmap

Structural patterns (e.g., "exactly two unique values") and Mixed/Composite rules proved to be the most difficult across all models. Simple "Order" rules were discovered in 60-80

3.6. Visualization of Search Dynamics

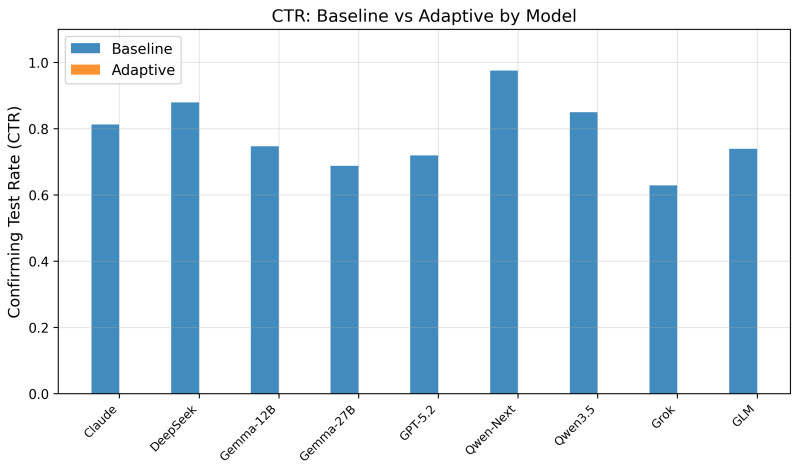

Figure 5: Confirming Test Rate (CTR): Reduction in Adaptive condition shows a shift toward falsification.

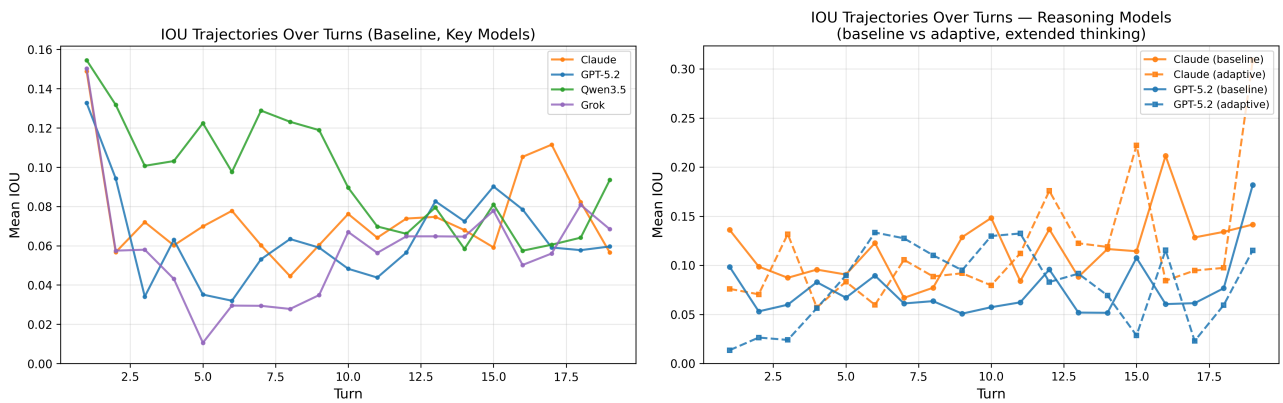

Figure 6: Left: Standard IOU trajectories (early peaking). Right: Reasoning model trajectories (sustained learning).

...

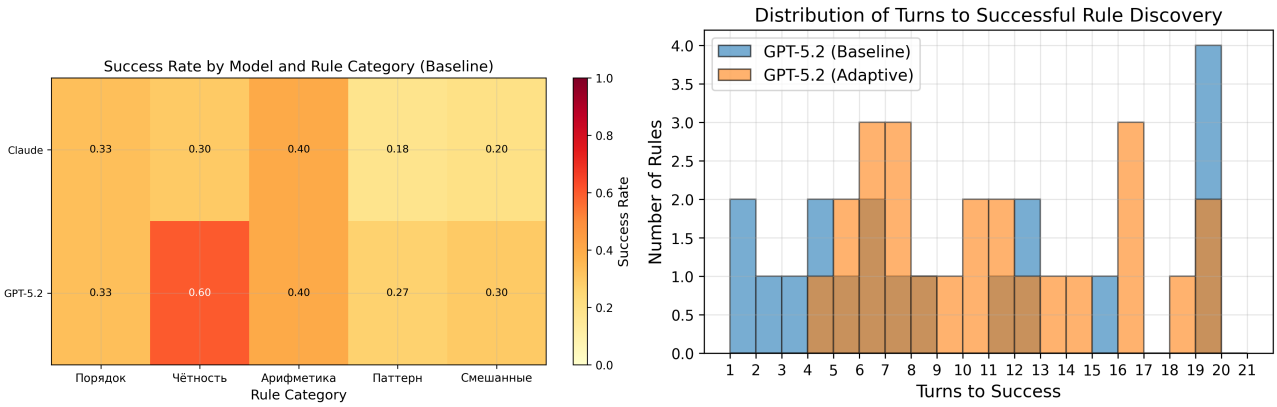

Figure 7: Left: SR Heatmap by Rule Category. Right: Average turns to success (Reasoning models are significantly faster).